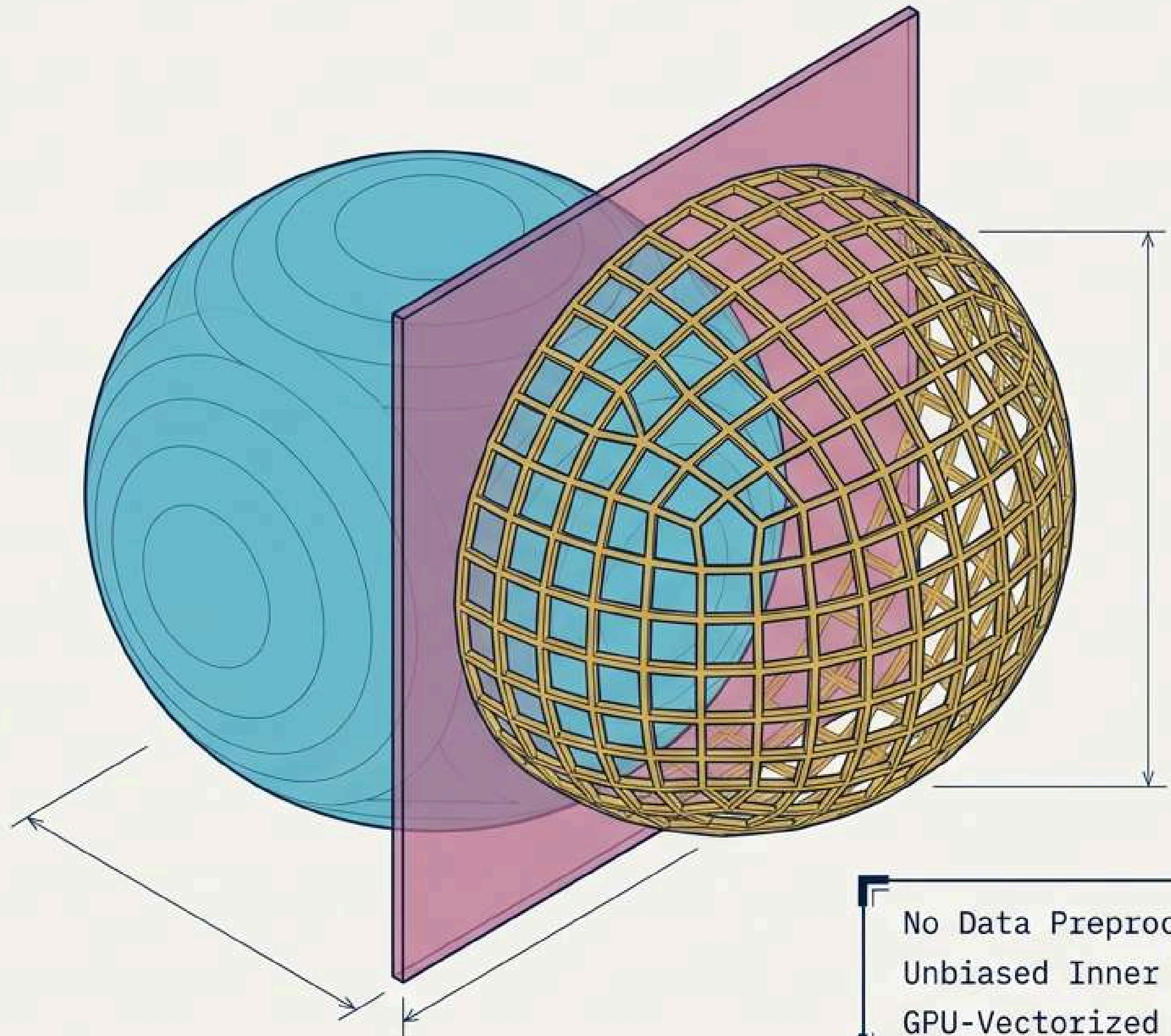


TurboQuant: Online Vector Quantization

Near-Optimal Distortion
Rates for LLM KV Caches
and Vector Search

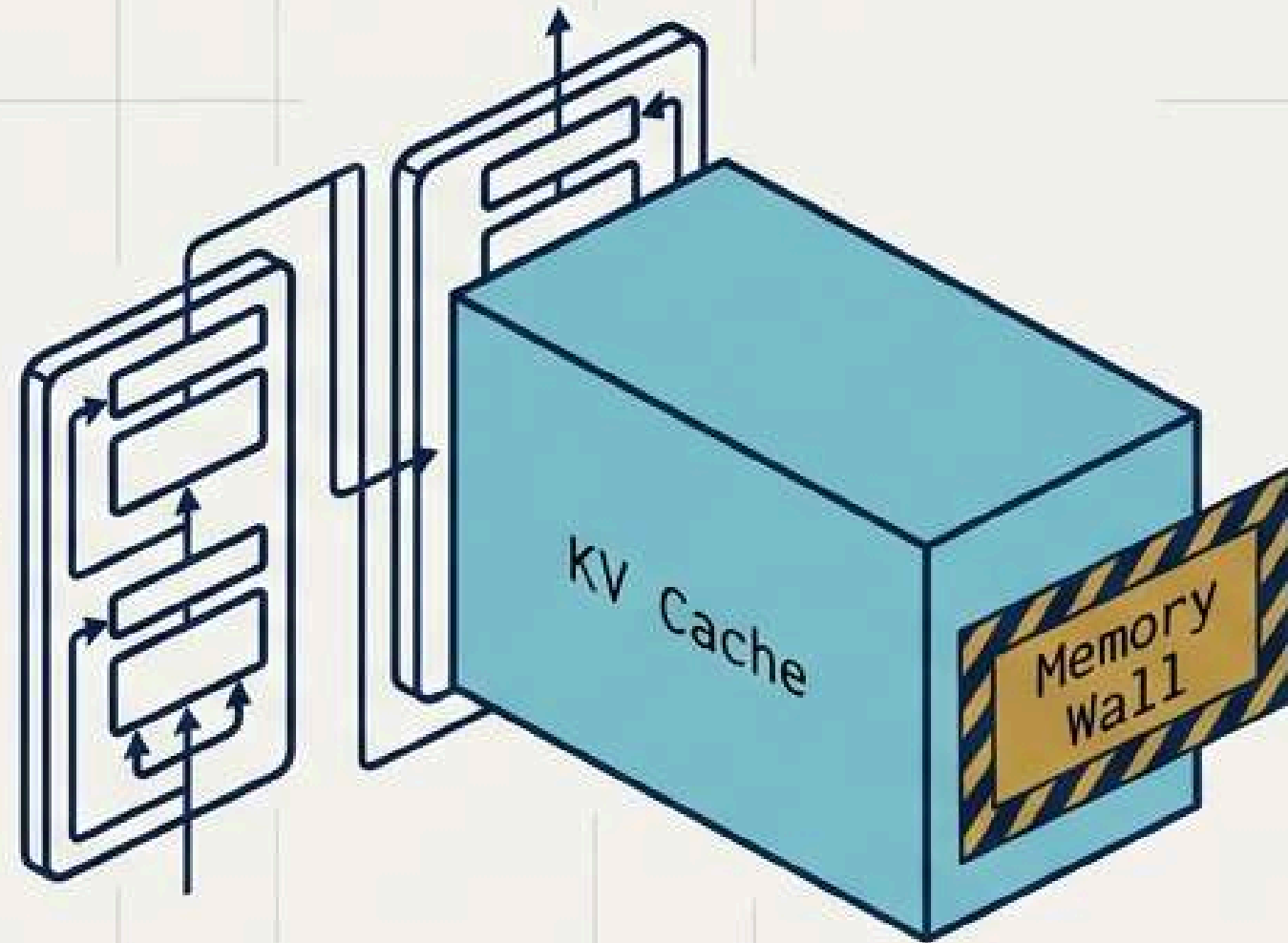
Presented By
Nizaul Rahman
2025MCS2099



No Data Preprocessing
Unbiased Inner Products
GPU-Vectorized

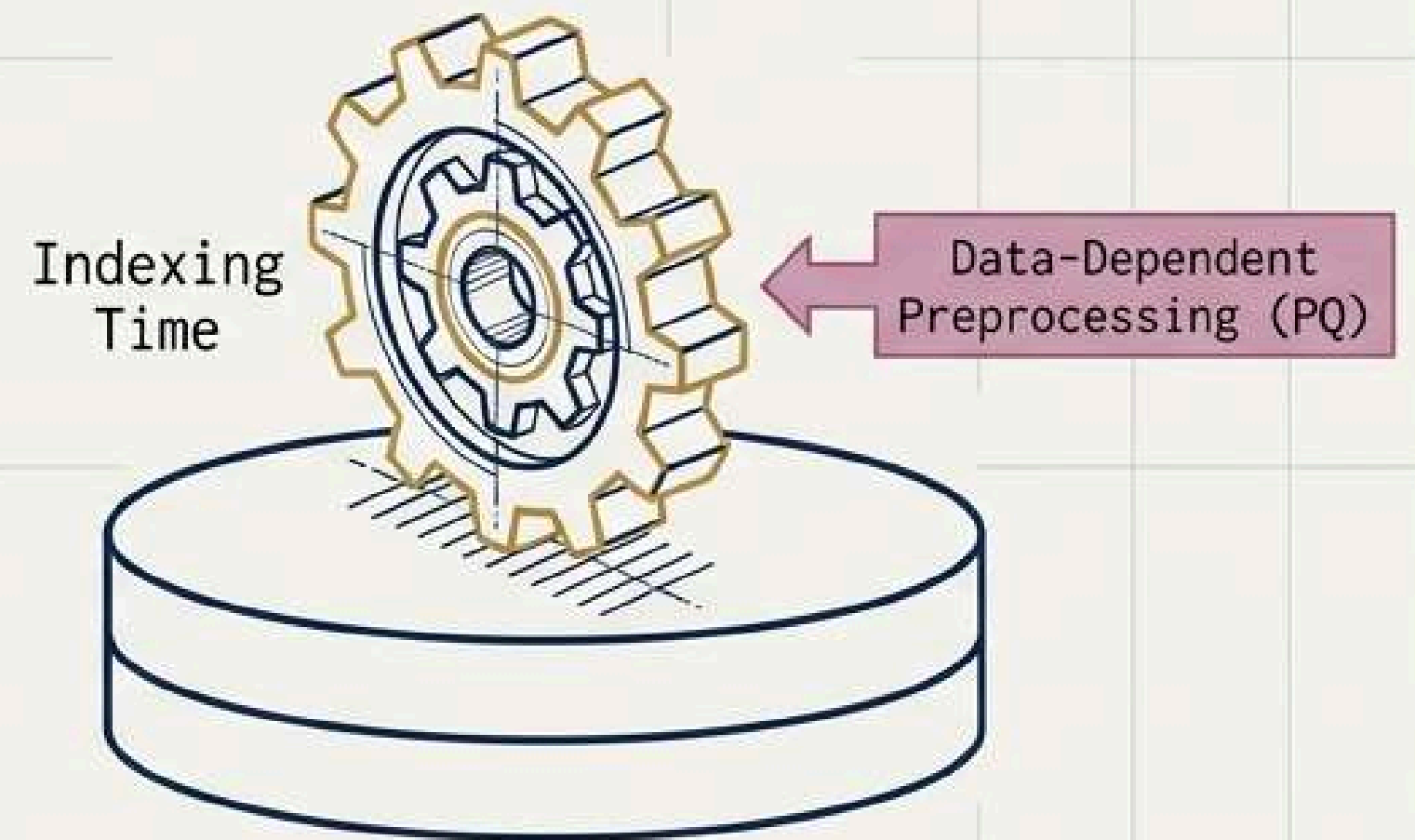
High-Dimensional Vectors are Choking Modern AI

LLM Bottleneck



Scaling context lengths exponentially inflates the Key-Value (KV) cache, crushing inference memory. Token eviction loses data.

Vector DB Bottleneck

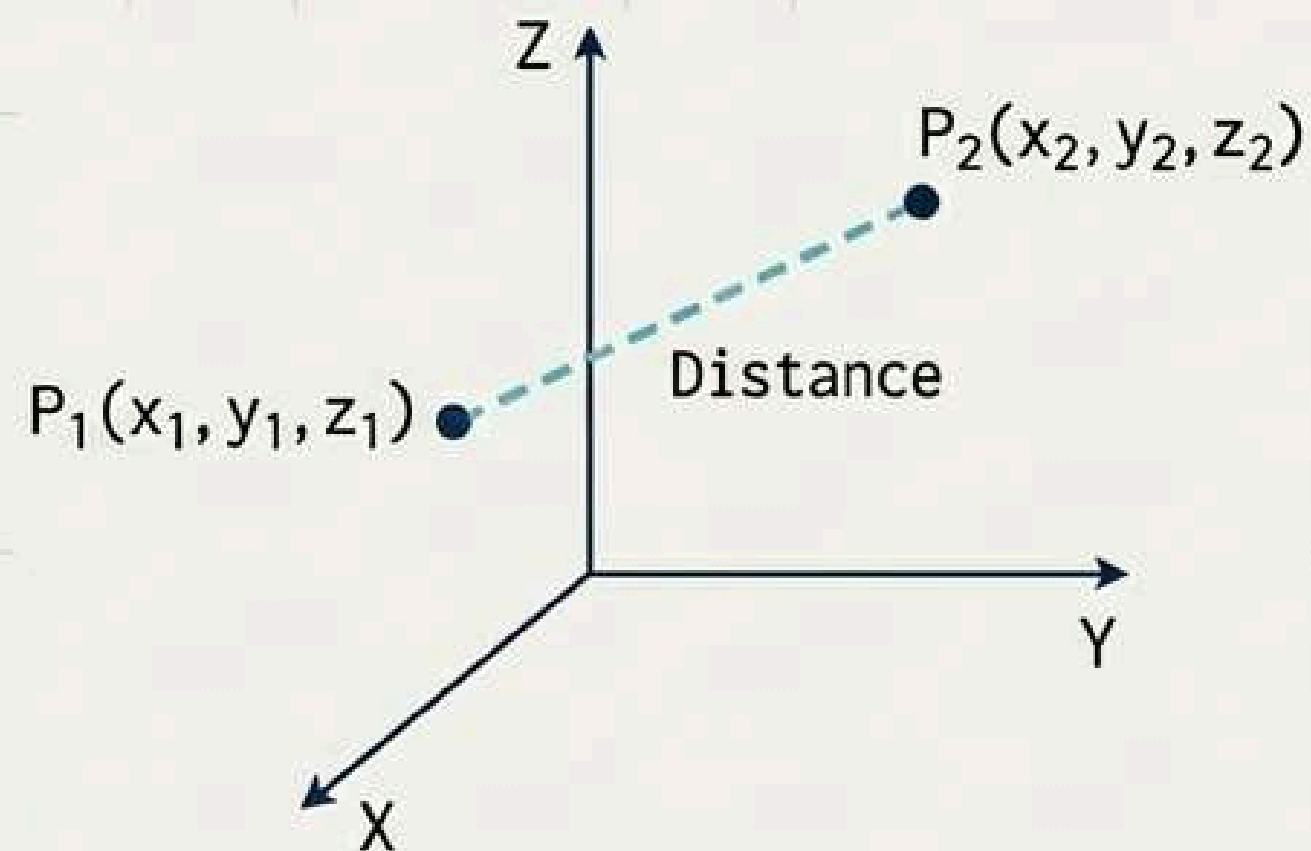


Existing Product Quantization (PQ) methods demand massive, slow k-means preprocessing.

The industry requires an online (instant, data-oblivious) quantizer that preserves true geometric fidelity.

The Mathematical Dual-Objective

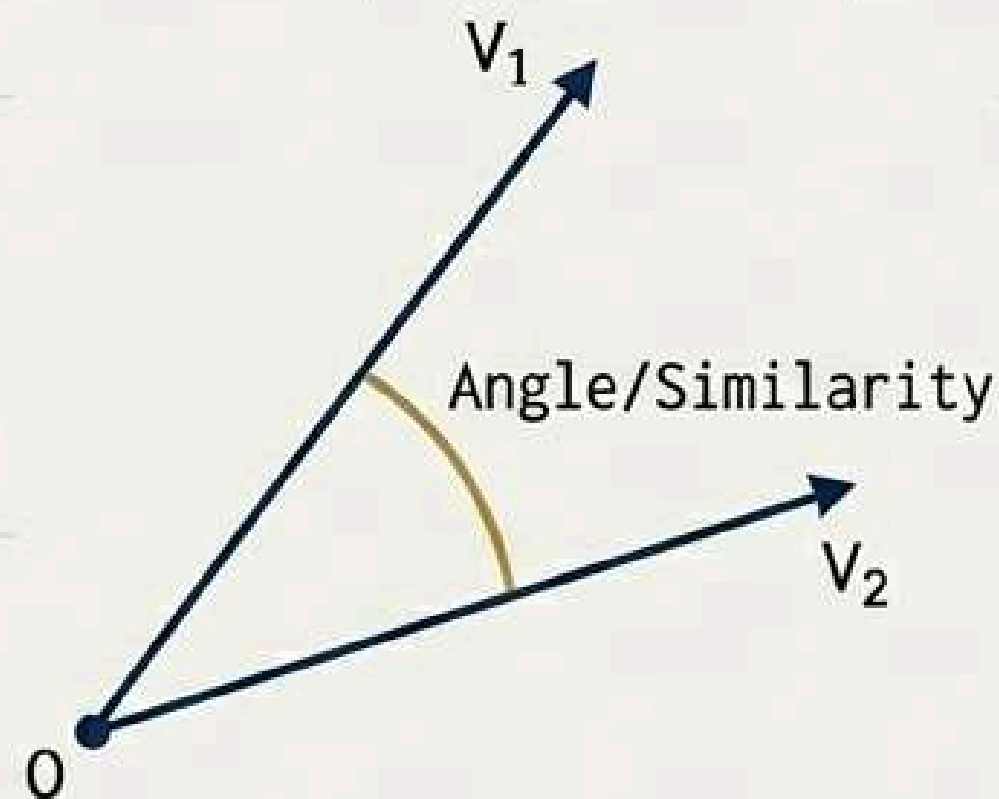
Minimizing Mean-Squared Error (MSE)



Goal: Preserve Euclidean distance.

$$D_{\text{mse}} := \mathbb{E}_Q \left[\|\mathbf{x} - \mathbf{Q}^{-1}(\mathbf{Q}(\mathbf{x}))\|_2^2 \right]$$

Preserving Inner Product Fidelity



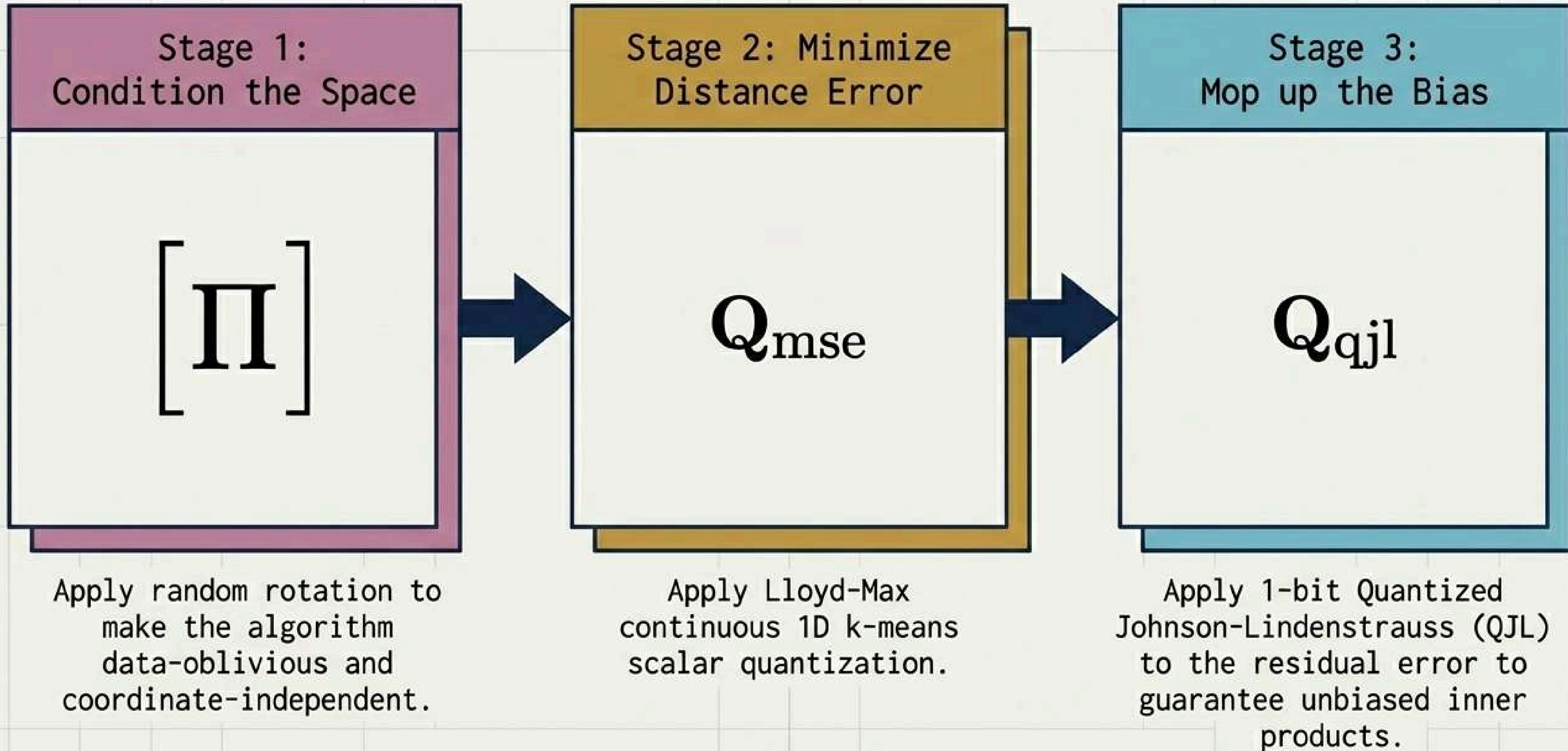
Goal: Preserve cosine similarity and angles.

$$D_{\text{prod}} := \mathbb{E}_Q \left[|\langle \mathbf{y}, \mathbf{x} \rangle - \langle \mathbf{y}, \mathbf{Q}^{-1}(\mathbf{Q}(\mathbf{x})) \rangle|^2 \right]$$

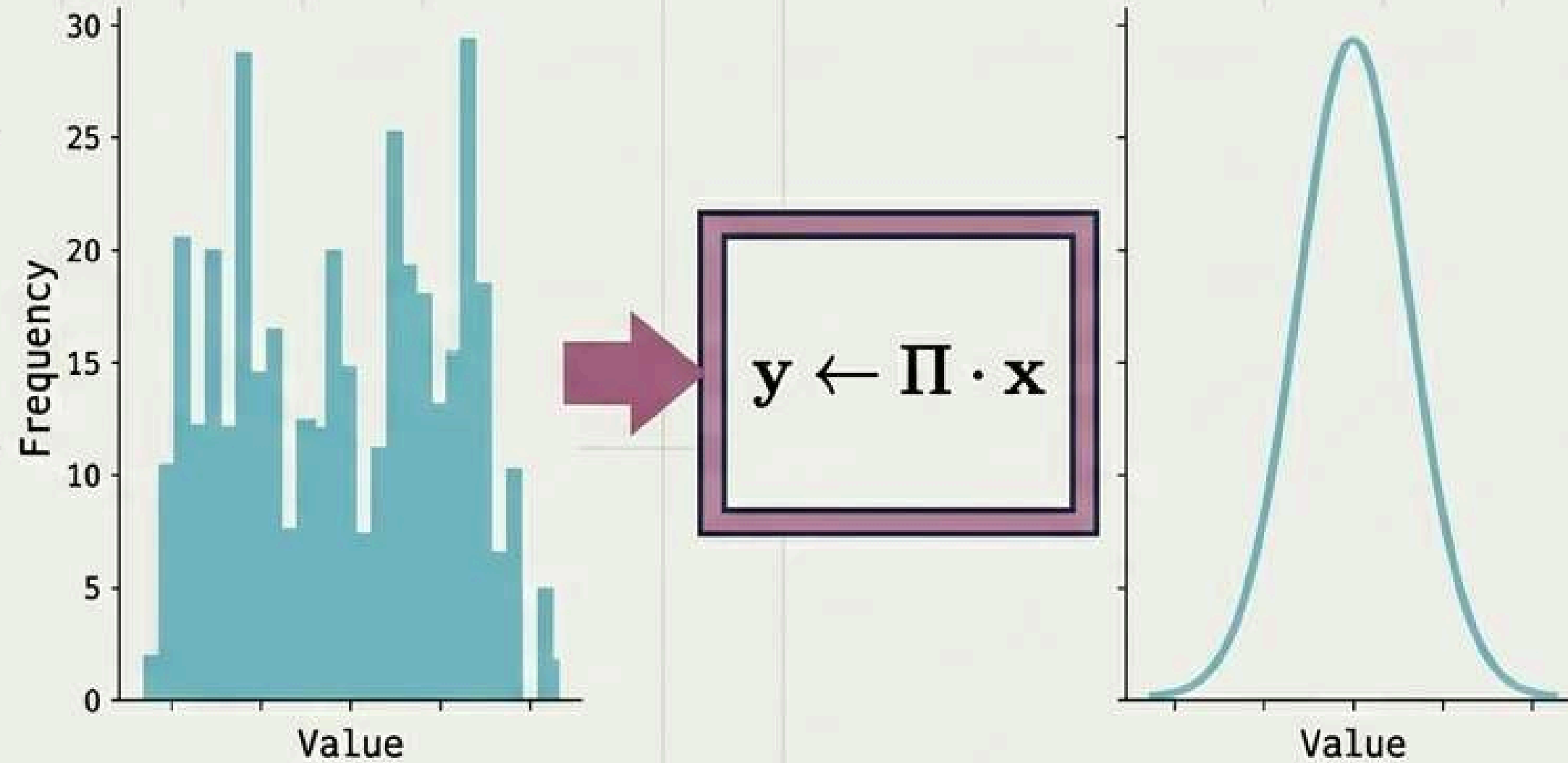
Strict Requirement: The estimator must be strictly unbiased:

$$\mathbb{E}_Q[\langle \mathbf{y}, \tilde{\mathbf{x}} \rangle] = \langle \mathbf{y}, \mathbf{x} \rangle$$

The TurboQuant Architecture



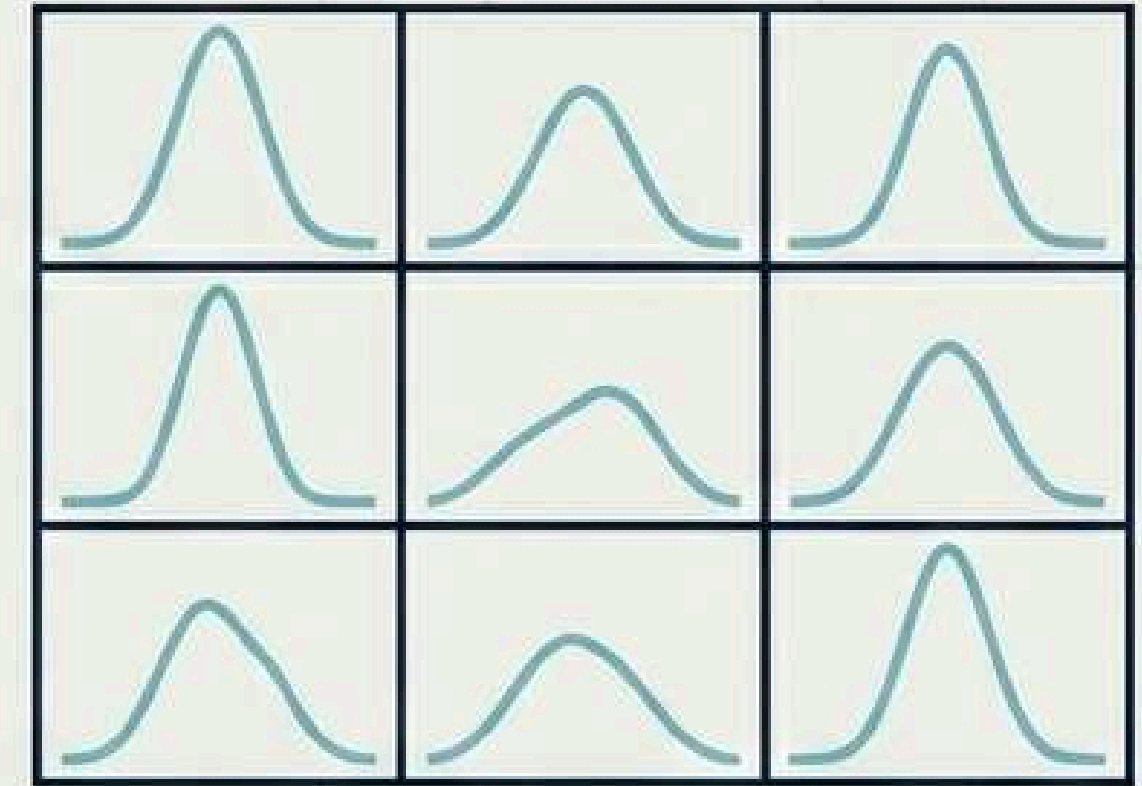
Stage 1: Random Rotation (Data-Oblivious Conditioning)



Constructing the Matrix: The rotation matrix Π in $\mathbb{R}^{(d \times d)}$ is generated instantly via QR decomposition of a random matrix with i.i.d. Normal entries $N(0,1)$.

The Geometric Magic:

By rotating the worst-case input vector x across a high-dimensional sphere, every coordinate y_j is forced into a Beta distribution converging to $N(0, 1/d)$.



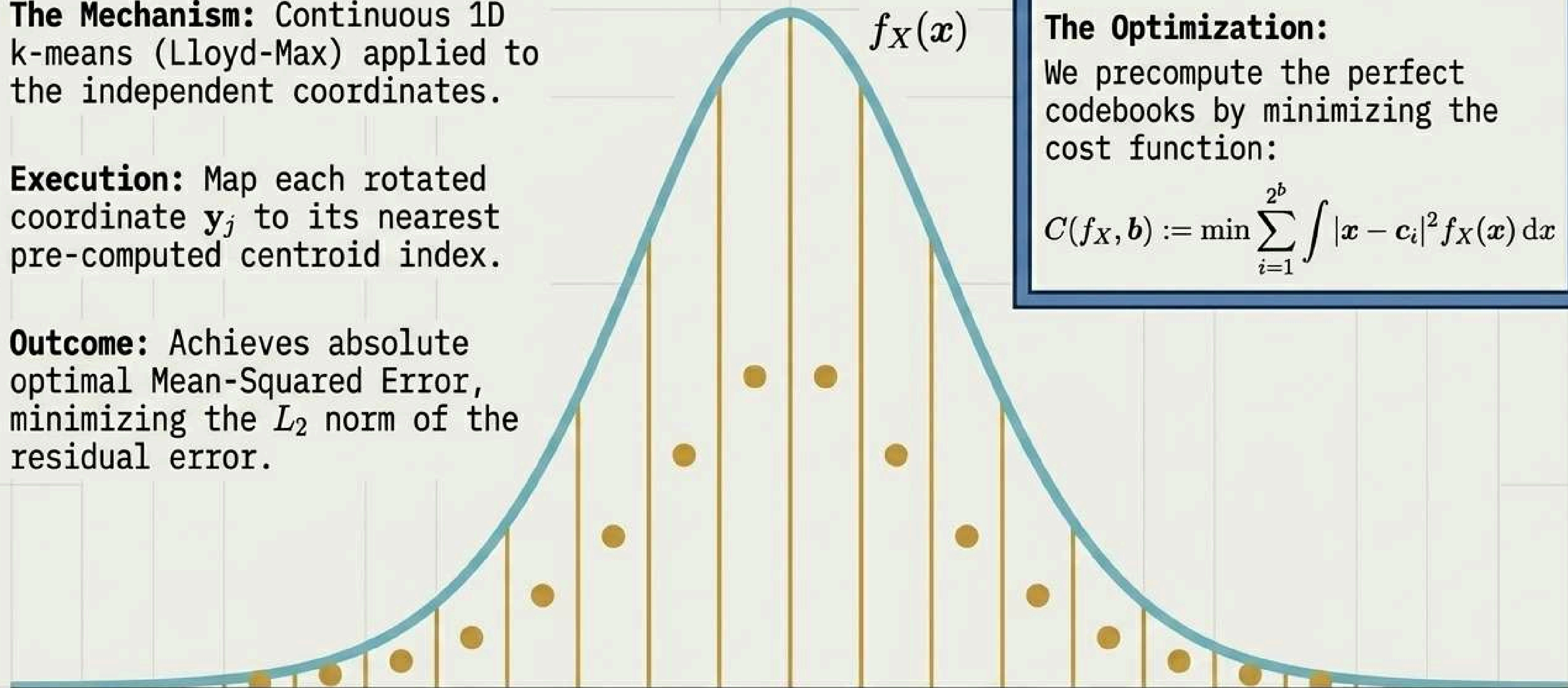
The Result: Distinct coordinates become nearly independent, allowing us to quantize them separately without complex joint-optimization.

Stage 2: Optimal MSE Quantization

The Mechanism: Continuous 1D k-means (Lloyd-Max) applied to the independent coordinates.

Execution: Map each rotated coordinate y_j to its nearest pre-computed centroid index.

Outcome: Achieves absolute optimal Mean-Squared Error, minimizing the L_2 norm of the residual error.

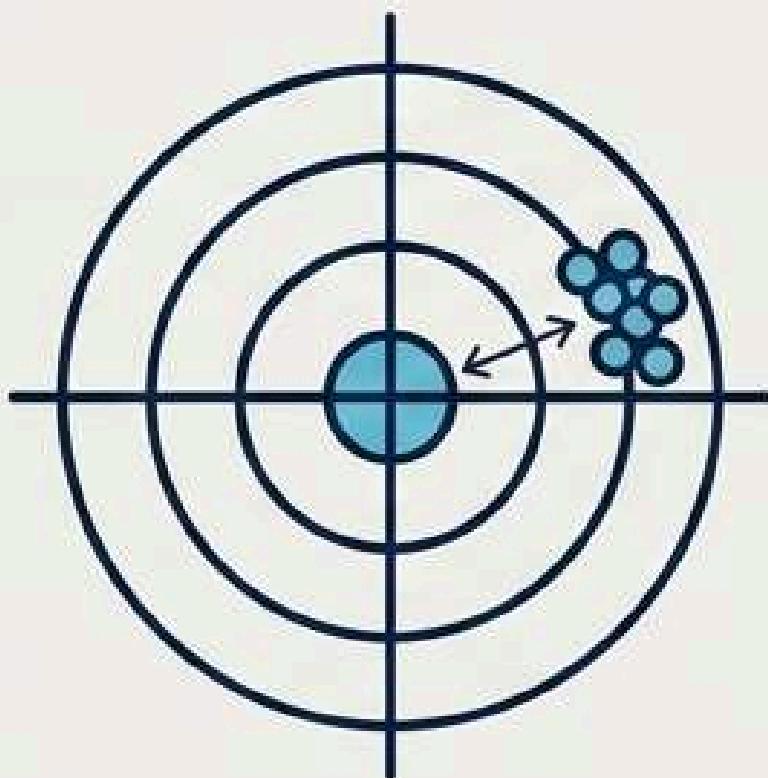


The Optimization:

We precompute the perfect codebooks by minimizing the cost function:

$$C(f_X, \mathbf{b}) := \min \sum_{i=1}^{2^b} \int |x - c_i|^2 f_X(x) dx$$

The Inner-Product Bias Flaw



The Problem: Optimal MSE quantizers are fundamentally biased for inner product estimation. At 1-bit widths, the multiplicative bias is a massive $2/\pi$.

The true vector

x

The Stage 2 MSE estimate

$\tilde{x}_{mse}$

The Residual

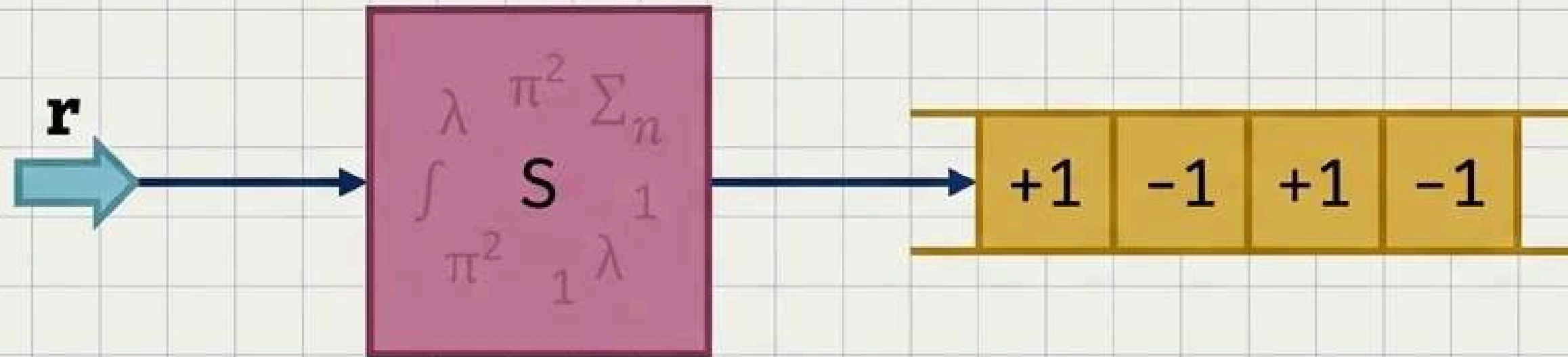
r

The Transition: We capture the exact mathematical difference between the true vector and our MSE estimate.

$$r = x - \tilde{x}_{mse}$$

The Hand-off: This residual vector has a tiny L_2 norm but holds the exact geometric data needed to correct the bias.

Stage 3: QJL Residual Quantization



The Mechanism:

Apply a 1-bit Quantized Johnson-Lindenstrauss (QJL) transform strictly to the residual $\mathbf{r}$.

The Cost:

Exactly 1 additional bit per coordinate.

The Formula:

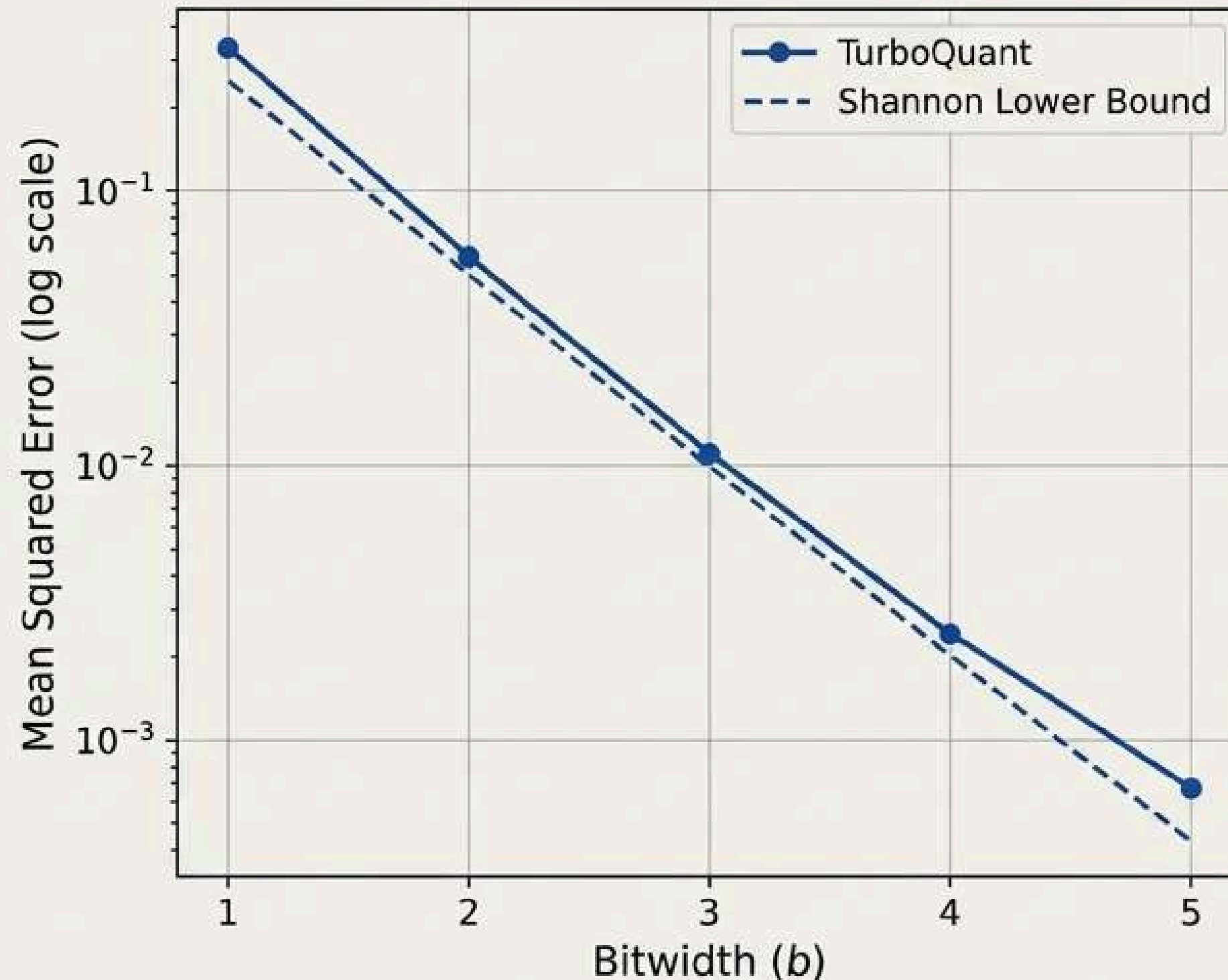
$$Q_{\text{qjl}}(\mathbf{r}) := \text{sign}(\mathbf{S} * \mathbf{r})$$

(where S in $\mathbb{R}^{d \times d}$ is an i.i.d $N(0,1)$ random matrix)

The Guarantee:

Combining the MSE estimate with this QJL estimate yields a mathematically unbiased inner product with an optimal error rate.

Theoretical Optimality: The Shannon Lower Bound



The Absolute Limit:

By Yao's minimax principle and the Shannon Lower Bound, the best possible MSE distortion achievable by any algorithm is $\geq 1 / 4^b$.

TurboQuant's Limit:

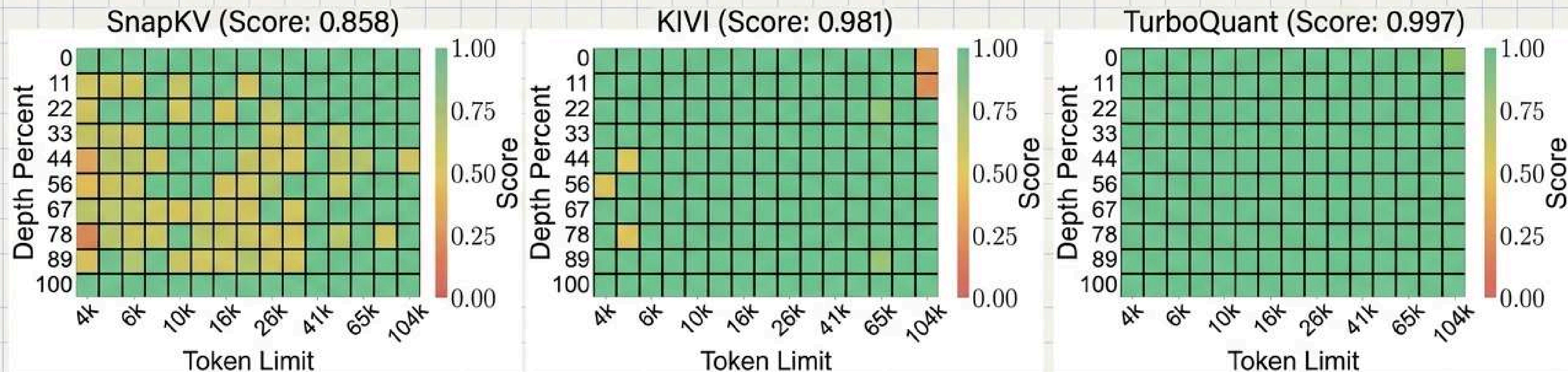
TurboQuant is mathematically proven to achieve $\leq (\sqrt{3\pi} / 2) * (1 / 4^b)$.

The Takeaway:

TurboQuant operates strictly within a tiny constant factor ($\sim 2.7x$) of absolute information-theoretic perfection across all bit-widths.

KV Cache Compression: Zero Retrieval Degradation

Test: Needle-In-A-Haystack (Llama-3.1-8B, 104k Context).



Competitor Failures:

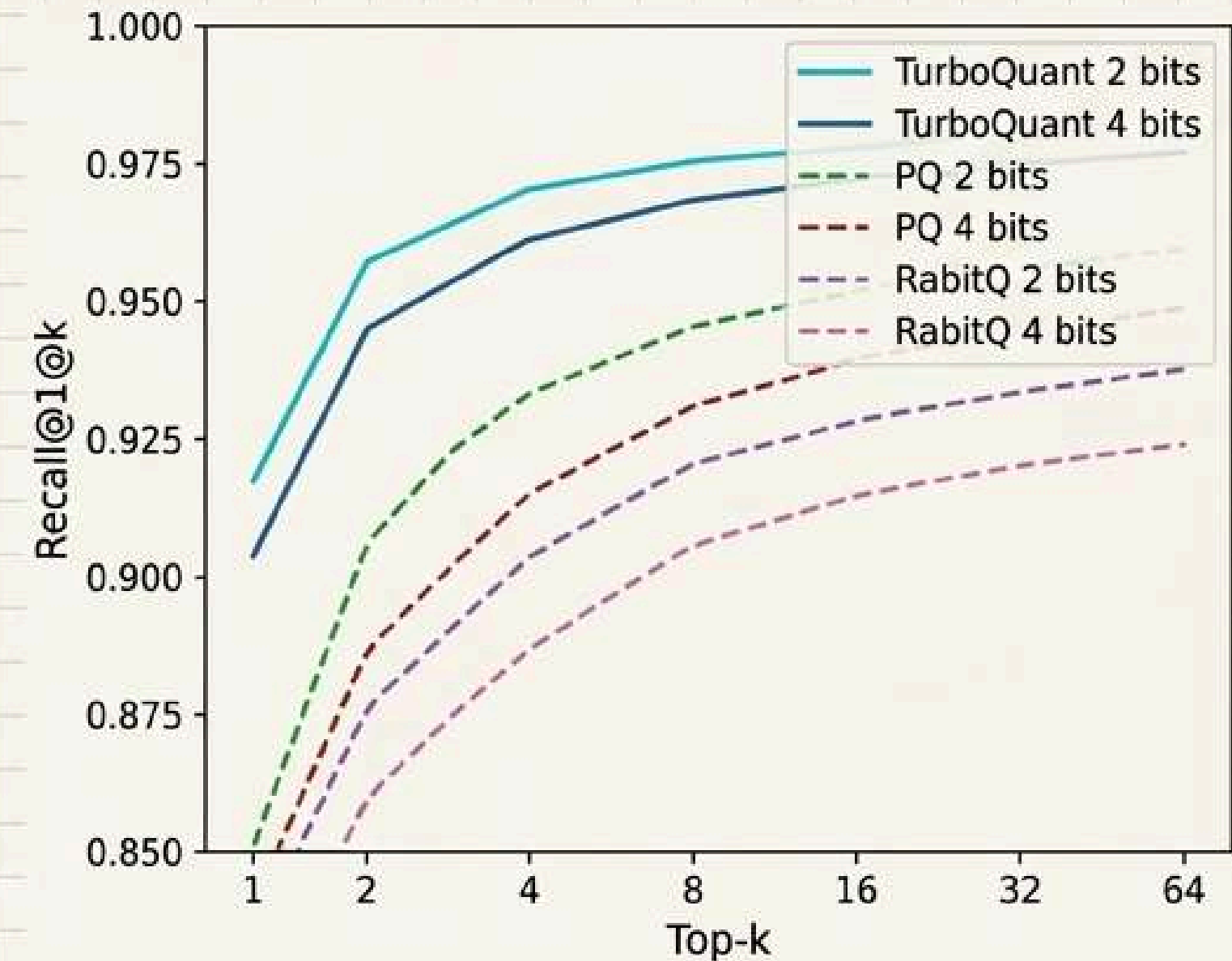
Token-level eviction (SnapKV) and scalar quantization (KIVI) destroy critical context.

TurboQuant Victory:

At an extreme 4x compression (2.5 to 3.5 bits/channel), TurboQuant achieves a perfect 0.997 recall score—identical to the uncompressed, 16-bit Full Precision model.

Vector Search: Eradicating Indexing Latency

Method	Hardware Vectorization	Data-Dependency	Indexing Time (d=1536)
Product Quantization (PQ)	Yes	High (Heavy k-means)	~239.75s
RabitQ	No (CPU only)	None	~2267.59s
TurboQuant (Ours)	Yes (GPU-Native)	Zero (Data-oblivious)	0.0013s



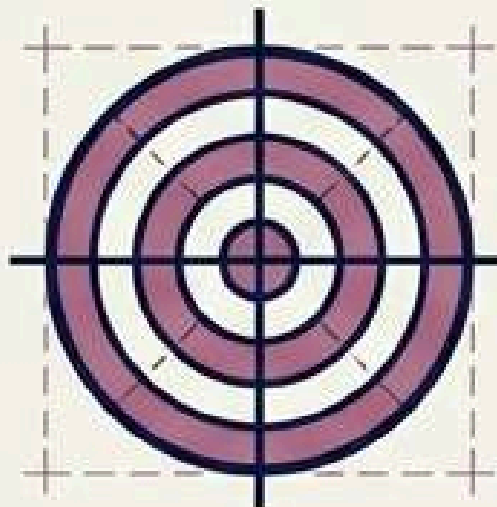
Conclusion: Orders of magnitude faster to index, with superior Recall@1 accuracy because geometry is mathematically preserved.

Summary: The TurboQuant Advantage



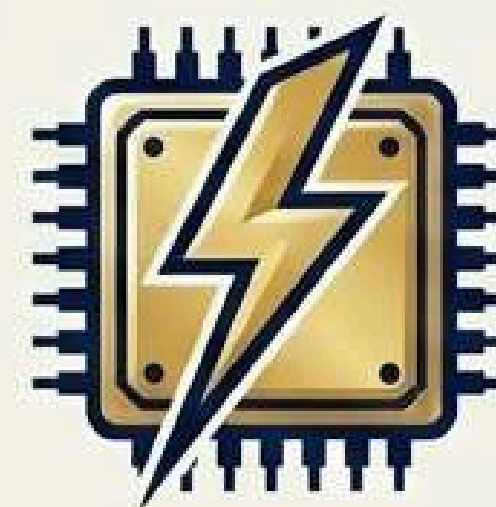
1. Fundamentally Sound

Anchored by Shannon's Source Coding theory. Distortion rates are mathematically guaranteed to be **near-optimal**, regardless of the data.



2. Unbiased by Design

The novel 2-stage architecture (Optimal MSE + 1-bit QJL Residual) completely eliminates the inner-product bias fatal to standard quantizers.



3. Deployment Ready

100% data-oblivious. Requires **zero preprocessing**.
Designed natively for **GPU** vectorization to drop into **LLM KV caches** and **Vector DBs** instantly.

Experiments & Results

Method	KV Size	SingleQA	MultiQA	Summarization	Few shot	Synthetic	Code	Average
Llama-3.1-8B-Instruct								
Full Cache	16	45.29	45.16	26.55	68.38	59.54	46.28	50.06
KIVI	3	43.38	37.99	27.16	68.38	59.50	44.68	48.50
KIVI	5	45.04	45.70	26.47	68.57	59.55	46.41	50.16
PolarQuant	3.9	45.18	44.48	26.23	68.25	60.07	45.24	49.78
TURBOQUANT (ours)	2.5	44.16	44.96	24.80	68.01	59.65	45.76	49.44
TURBOQUANT (ours)	3.5	45.01	45.31	26.00	68.63	59.95	46.17	50.06
Ministral-7B-Instruct								
Full Cache	16	47.53	49.06	26.09	66.83	53.50	47.90	49.89
TURBOQUANT (ours)	2.5	48.38	49.22	24.91	66.69	53.17	46.83	49.62

Table 1: LongBench-V1 [10] results of various KV cache compression methods on Llama-3.1-8B-Instruct.

Approach	d=200	d=1536	d=3072
Product Quantization	37.04	239.75	494.42
RabitQ	597.25	2267.59	3957.19
TURBOQUANT	0.0007	0.0013	0.0021

Table 2: Quantization time (in seconds) for different approaches across various dimensions using 4-bit quantization.

Experiments & Results

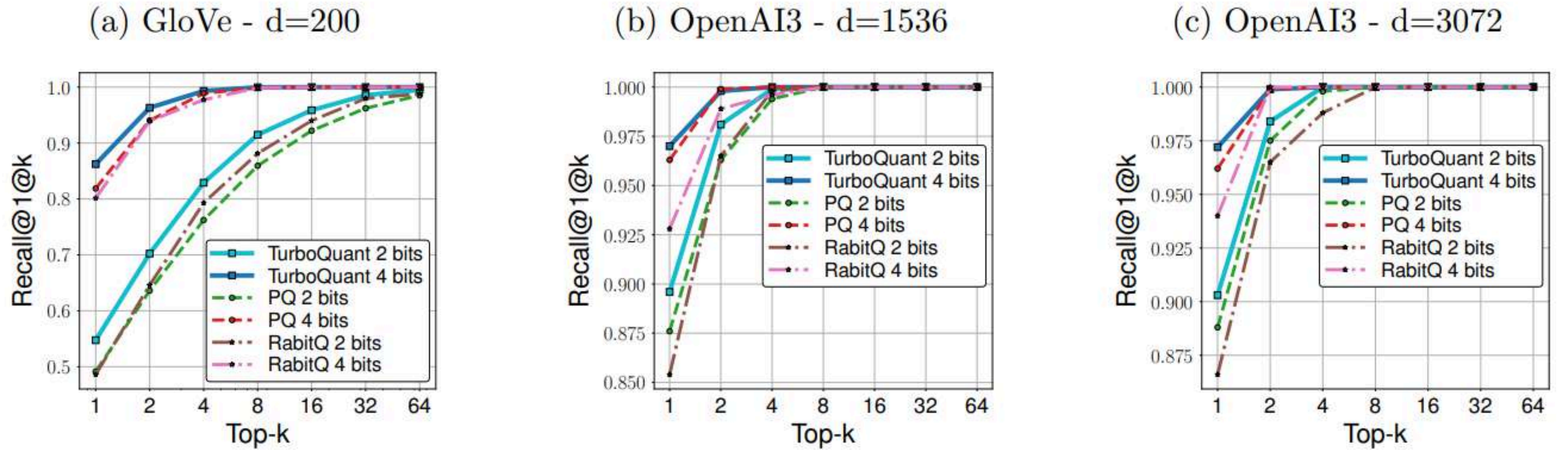


Figure 5: Recall comparison on different datasets with different embedding dimensions.